Supplementary Methods: From Fuel Customers to Flight Geography

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Source-neutral flight-stage contract

All provider-specific adapters terminate at one validated record. Required fields are a stable flight identifier, fiscal year, origin and destination ICAO codes, domestic/international boundary, activity count, record granularity, source tier, operator, aircraft type, trajectory quality, imputation method and source reference. Optional fields include date and timestamps, callsign, ICAO24, registration, engine, passengers, cargo, great-circle and trajectory distance, and taxi, take-off, climb, cruise, descent and landing fuel. Phase fuel must sum to total fuel; the geography implied by endpoints must agree with the stored boundary.

Four tiers are enforced. Tier A is an observed flight and trajectory; Tier B is an observed flight stage with a synthetic trajectory; Tier C is a reconstructed annual route stratum; Tier D is aggregate residual activity. Tiers A and B require flight-stage granularity and timestamps; Tier C cannot contain invented dates; Tier D cannot be assigned fictional endpoints. ADSB.lol, OpenSky (ADSB.lol, 2024; Schäfer et al., 2014), a future PCAA operational export and a canonical CSV/Parquet import therefore feed the same emissions and accounting code.

2. Provisional route and aircraft reconstruction

The FY2023–24 route graph contains only routes with contemporaneous timetable, airline-report or route-announcement evidence. Each evidence record carries applicable dates, operator, source class and URL. An iterative constrained allocation balances airport origin and destination margins. A joint linear allocation then assigns operator departures without exceeding either an airline control or route capacity. Airport, route and operator deficits are retained as Tier D.

Let x_{aij} denote annual stages by operator a from airport i to j. The central matrix minimises deviation from documented seed weights subject to airport outflow and inflow margins, operator departure controls, reciprocal-route support and non-negativity. Over the same feasible set, separate programmes minimise and maximise $\sum_{aij} x_{aij} d_{ij}$, where d_{ij} is great-circle distance. The resulting matrices isolate route-graph uncertainty without changing activity totals or inventing unsupported links.

Aircraft are assigned as probabilities by airline and route, not as invented registrations. OpenAP (Sun et al., 2020) provides synthetic phase profiles for supported types. Unsupported types remain visibly unfuelled in the native model and receive a separate ICAO calculator imputation only in the coverage-completion experiment (ICAO, 2024). Low, central and high payload configurations change dispatch mass through passengers, cargo and mail. Detour, taxi time and weather are varied separately in the Monte Carlo so their effects are identifiable; the main paper reports the central common-support comparison.

The installed OpenAP deterministic generator documents cruise range in kilometres but compares the supplied value with an internally metre-scaled trajectory. The adapter therefore identifies the installed convention with a bounded probe, selects the highest feasible cruise altitude on short routes and requires generated horizontal distance to match the detoured route within 3% or 5 km. It also requires profile duration to increase materially with route length and each phase-fuel sum to equal total fuel. These invariants detected the earlier near-route-invariant profiles and remain executable regression checks.

3. PCAA payload and departure controls

Central payload is no longer defined by one generic MTOW fraction. For each operator and boundary, PCAA embarked passengers are divided by published departures; assigned aircraft- family shares supply weighted seats; loaded cargo and mail are divided by the same departure control. Table 1 reports the domestic values used by M2. Low and high cases use 62.5% and 87.5% passenger loads unless the observed centre lies beyond those screening values; this

guarantees low ≤ central ≤ high up to physical seat capacity. The outputs retain a warning rather than silently treating an impossible load as valid.

Operator	Pax/dep.	Seats	Implied load	Cargo+mail kg/dep.
Airblue (ABQ)	170.4	194	0.878	0
Fly Jinnah (FJL)	156.8	180	0.871	811
PIA	109.5	147	0.745	723
SereneAir (SEP)	173.1	189	0.916	2187
AirSial (SIF)	174.4	180	0.969	1308

Table 1: Domestic operator-average payload controls. They constrain, but do not claim to observe, route-specific payload.

The airline report contains 19,752 domestic departures and arrivals, versus 20,300 stages from airport movements divided by two (difference −548, or −2.7%). Internationally, 45,723 published departures compare with 47,633.5 half-movements (difference $-1,910.5$, or −4.0%); departures exceed arrivals by 526. The incomplete international reconstruction assigns only 2,510 of PIA's 9,009 departures and therefore produces an above-capacity warning if its partial aircraft mix is applied to all reported passengers. That warning is retained as a coverage diagnostic and is not used to alter the domestic payload model.

4. Extended reconciliation diagnostics

Figure 1 retains the full graphical source-row bridge. The main paper reports the same values in a compact conservation table so that the empirical argument is not interrupted by every structural destination.

The monthly and extended annual comparisons in Figure 2 show why the latest annual ratios are treated as evidence rather than a permanent identity. Monthly combined ratios fluctuate around unity; over the full complete OECD overlap, combined closure improves while resident and foreign discrepancies retain opposite signs.

5. Comparison independence and model diagnostics

Table 2 gives the full comparison registry underlying the compact source-role table in the main paper. Calibration, external provenance and statistical independence are separate attributes.

6. Observed-data adapters and activation gate

The implemented OpenSky adapter queries flight records first and state vectors only for selected flight identifiers and time bounds. The ADSB.lol adapter instead streams ODbL daily readsb archives, honours the trace new-leg bit, links only low-altitude airport candidates and emits the same Tier-A records without redistributing the underlying trace database (readsb contributors, 2026). Coverage is evaluated against PCAA airport and airline controls by month, boundary and operator. An observed ledger may replace the calibrated OECD resident share only when domestic and international route and fuel coverage each exceed 90% for both OCAC resident customer groups. Below that gate, modelled fuel is a partial diagnostic and uncovered OCAC fuel remains residual. This rule prevents a high-coverage domestic sample and low-coverage international sample from producing a biased customer split.

7. ADSB.lol coverage-pilot result

The selected dates are 15 August 2023, 15 November 2023, 15 February 2024 and 15 May 2024. Their release assets total 7.05 GiB. Streaming 219,596 trace members yields 1,543 aircraft traces and 4,281 legs in the broad regional candidate rectangle, but zero low-altitude links to a Pakistani airport and therefore zero canonical Tier-A stages. One malformed member per archive is counted as a parse failure. The nearest horizontal candidate is 0.25 km from a catalogued airport; it is rejected by the altitude-relative gate. Only these aggregate counts, release identifiers, file sizes, licence and method are distributed. The raw trace cache and a derived reusable flight database are withheld pending ODbL derivative-database review.

Two positive controls use the same canonical-stage code. A synthetic OPKC–OPLA trace emits one Tier-A domestic stage with the expected endpoints. The unmodified 15 August archive, scanned with the identical 35-km and elevation-relative gate around OMDB, emits 700 airport-linked legs; 686 have callsigns and 697 have aircraft types. Both controls pass. The zero Pakistan result is therefore evidence about this sample's coverage, not a silent parser or endpoint-gate failure.

8. Official-reporting extraction

The official CRT ZIP is read in memory and the energy workbook is selected by exact filename. The extractor takes emissions from Table 1, domestic activity from Table 1.A(a)s3 and bunker activity plus the domestic/international distribution from Table 1.D. It asserts that both aviation categories, both activity cells and the distribution exist. The output retains the ZIP SHA-256, reported fuel label and jet-kerosene notation. BTR/NC3 narrative entries are a separate normalized control because their rounded printed values cannot be treated as CRT-precision observations.

The archived file is `PAK_CRT_2024.zip`. Its SHA-256 is printed as two consecutive 32-character blocks:

```
afba70d10be0694aea6b2ba2cec311e1
49473346daa206319138185c190e8f38
```

The selected workbook is `ENERGY-PAK-CRT.xlsx`; its two consecutive SHA-256 blocks are:

```
ca504f7eb0894573d790b1b4c08576d4
fbe795acd56ef5e5c4120b0d3897a944
```

Table 3 reproduces the cells carrying the central interpretation. The generated cell-audit CSV contains all 21 pinned cells, including CH4, N2O and CO_2e, along with raw data type, formula, unit, transformation, transformed value and notation key.

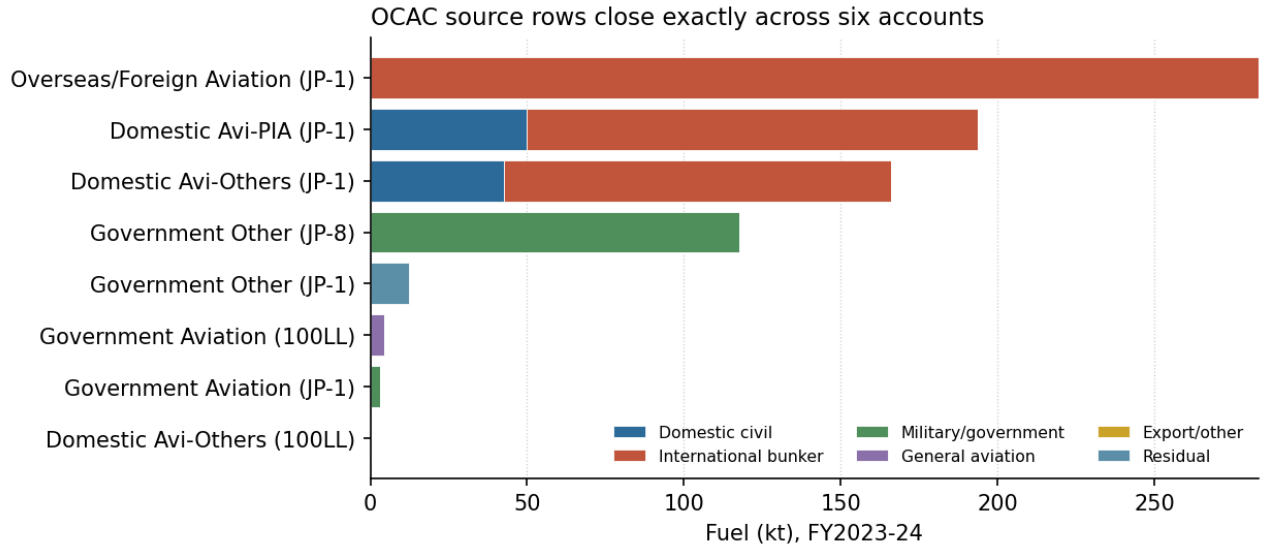


Figure 1: FY2023–24 OCAC rows assigned across six non-overlapping accounts under the OECD-calibrated resident share.

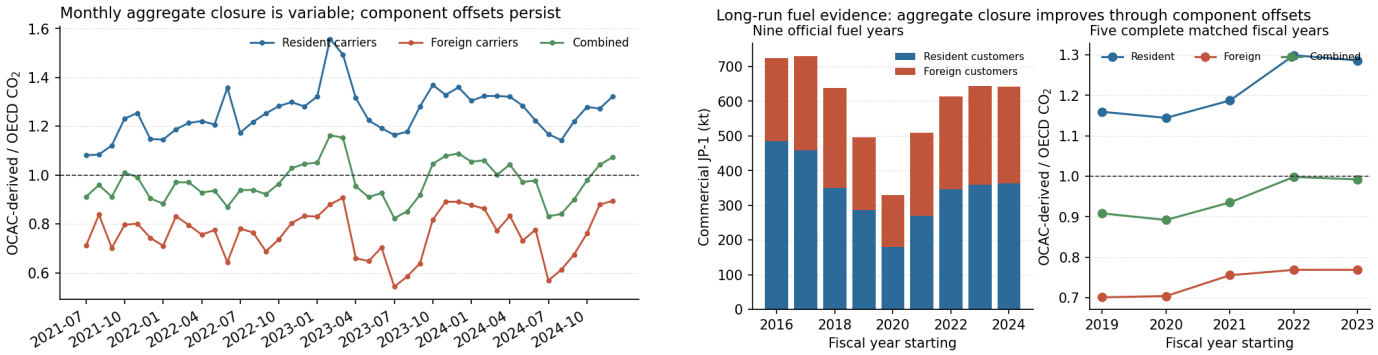


Figure 2: Supporting temporal evidence. (a) Monthly OCAC-derived to OECD CO_2 ratios. (b) Nine annual fuel accounts and five complete fiscal-year component comparisons.

Comparison	Purpose	Shared input	Independence	Calibration
Movement vs OECD domestic	External territory comparison	Possible ICAO performance conventions	Partial	No
Fuel bridge vs OECD domestic	Partition consistency	OECD resident share used directly	None	Yes
OCAC total vs OECD combined	Gross-boundary closure	Carbon conversion only	Large	No
Pakistan CRT vs OECD	Official-boundary diagnostic	Inventory year only	Large	No
PCAA airports vs Climate TRACE	Spatial corroboration	Possible surveillance/performance inputs	Partial	No
ADSB.lol vs PCAA daily rate	Coverage pilot	Airport coordinates only	Large but incomplete	No
ERA5 vs OpenAP still air	Weather sensitivity	Same synthetic route/fuel	Not validation	No
PCAA movements vs passengers	Movement-divisor check	PCAA reporting system	Targeted	No

Table 2: Complete independence and shared-input registry.

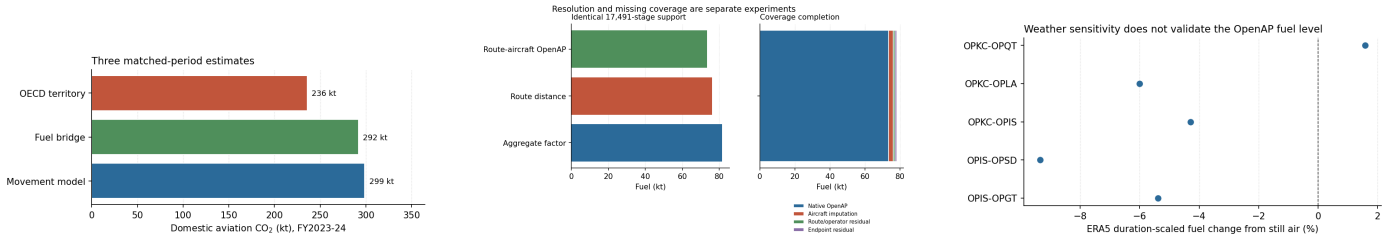


Figure 3: Implementation diagnostics moved from the main narrative. (a) Calibrated bridge, aggregate movement and OECD domestic comparison. (b) Common-support and coverage-completion ladders. (c) ERA5 duration sensitivity for representative routes.

Cell	Field	Raw value	Unit	Transformation
Table1!C25	Domestic CO_2	654.065328	kt CO_2	×1000 to t
Table1!C53	International CO_2	163.516332	kt CO_2	×1000 to t
Table1.A(a)s3!C17	Domestic aviation gasoline	9343.7904	TJ	Retain
Table1.A(a)s3!H17	Domestic gasoline CO_2	654.065328	kt CO_2	×1000 to t
Table1.A(a)s3!C18/H18	Domestic jet kerosene	NE / NE	Notation key	Retain NE
Table1.D!C12	International aviation gasoline	2335.9476	TJ	Retain
Table1.D!G12	International gasoline CO_2	163.516332	kt CO_2	×1000 to t
Table1.D!C11/G11	International jet kerosene	NE / NE	Notation key	Retain NE
Table1.D!L9	Domestic distribution	80.00000000000001	percent	Retain exact
Table1.D!M9	International distribution	19.999999999999986	percent	Retain exact

Table 3: Reproduced extract from the official CRT workbook. No value was manually transcribed into the normalized inventory: the extractor reads and asserts these cells.

9. ERA5 weather-duration audit

The reproducible CDS request selects temperature and horizontal winds at 250, 500 and 850 hPa over 23–38°N, 60–79°E at 12 UTC on the 15th of each FY2023–24 month. Bilinear interpolation samples 21 equally spaced points on six representative routes. At 250 hPa, wind projected on route bearing adjusts airborne duration; taxi fuel is unchanged and remaining OpenAP fuel scales with duration. Across 18 central route–aircraft strata the mean change is −3.90% (range −9.35% to +1.59%), comparable to the corrected 3.5% common-support M1–M2 difference. It does not model dispatch mass, reserves or wind-optimal routing and does not rerun the OpenAP equations with meteorological state.

10. Airport Air-Pollutant Layer

NOx, CO and HC are recalculated from representative aircraft–engine combinations in the ICAO Engine Emissions Databank (ICAO and EASA, 2026). The mapping is complete for the commercial fleet, but every row is marked imputed because aircraft registrations and actual engine fits are not available. The turboprop fallback and fuel-based SO2, PM2.5 and black-carbon factors are reported separately (EMEP/EEA, 2023).

These remain LTO mass estimates rather than a complete airport micro-inventory. Standard fleet shares stand in for airport-specific aircraft assignments, and actual taxi time, APU, ground support, brake and tyre sources are not estimated. Consequently the layer supports screening and method development, not claims about population exposure. A defensible airport inventory will require registration-to-engine matching and observed times in mode as recommended by ICAO Doc 9889 (ICAO, 2023).

The annual national core extends this LTO calculation through FY2024–25 using own-year airport movements and explicitly carried reference-era fleet composition. Unspeciated HC is retained as HC and does not fill the core NMVOC cell. Airport locations support LTO release geography; annual cruise mass remains unallocated below its supported administrative level, without synthetic route or altitude placement.

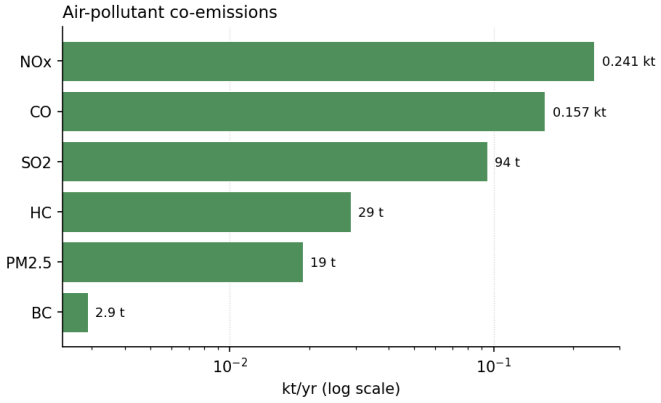


Figure 4: Domestic main-engine LTO pollutants on a logarithmic mass axis. Values use representative, explicitly imputed aircraft–engine mappings.

11. Products intentionally withheld without observed trajectories

No route-altitude grid, physical emissions-over-Pakistan product, observed taxi-time inventory, registration-to-engine inventory or airport micro-inventory is generated from Tier C records. Once Tier A/B records are available, the existing product layer can emit airport LTO, climb, cruise and descent segments, altitude bins and monthly gridded fields. National UNFCCC domestic, international bunker, resident-carrier and physical-airspace accounts remain separate products.

12. Fallback coverage

The documented graph allocates 20,097.5 of 20,300 domestic stages. Named domestic-airline assignment covers 19,752 departures. Native OpenAP fuel profiles cover 17,491.2 stages. The resident-carrier international reconstruction covers 4,653 of 16,701 departures (27.9%), so it does not pass the activation gate. These are coverage diagnostics, not claims that reconstructed records are observed flights.

13. Reproduction

```
pip install -e '.[aviation,gis]'
make aviation-study
make figures sync-figures
make test
```

The study target rebuilds official controls, the provisional fallback and every reconciliation product, and writes the tables below. Each is named: the paragraph this replaces described them by size—"42 monthly comparison rows", "240 monthly source rows"—which told a reader how large each output is without saying which file it is.

- `aviation_ocac_monthly.csv`: 240 monthly source rows: reported tonnes by customer category and fuel type, each with its source URL, local file, SHA-256 hash and revision note.
- `aviation_ocac_cumulative_monthly.csv`: the same source on a cumulative basis.
- `aviation_ocac_annual_summary.csv`: 9 annual OCAC summaries: resident commercial, foreign commercial and government JP-1 by fiscal year.
- `aviation_monthly_reconciliation.csv`: 42 monthly comparison rows, OCAC against OECD CO_2 with the resident and foreign ratios.
- `aviation_ocac_oecd_annual_comparison.csv`: the annual OCAC–OECD comparison, of which 5 years are complete: the others match six months each and are marked partial rather than dropped.
- `aviation_annual_component_comparison.csv`: the same comparison by component, with the role each component plays in it and the government JP-1 that is excluded.
- `aviation_bridge_source_rows_all_scenarios.csv`: every source row under every scenario, with its IPCC domestic, international-bunker, military 1A5b and general-aviation assignment.
- `aviation_fuel_bridge_fy2023_24.csv`: the reference year's fuel bridge on the same assignments.
- `aviation_reconciliation_fy2023_24.csv`: the reference-year reconciliation: domestic, international and residual fuel against the bottom-up domestic estimate.
- `aviation_model_resolution_fy2023_24.csv`: how much of the domestic flight population each model resolves—stages modelled against stages total, with the resulting coverage share.
- `aviation_common_support_ladder.csv`: the common-support ladder, each step's stage and fuel coverage under one activity definition.
- `aviation_two_year_replication.csv`: the two-year replication, including the route coverage share and the network assumption behind it.
- `aviation_timeseries_2006_fy2023_24.csv`: domestic LTO cycles and CO_2e by fiscal year from 2006.
- `aviation_uncertainty_inputs_complete.csv`: the complete input register: central value, 95 % range, distribution, applicable model, correlation group and uncertainty class per variable.
- `aviation_m2_uncertainty_intervals.csv`: the intervals by model, route graph and completion method, with the draw count and seed.
- `aviation_m2_uncertainty_contributions.csv`: what each uncertainty group contributes, as the variance reduction from freezing it.
- `aviation_uncertainty_sensitivity.csv`: the one-at-a-time sensitivity: each variable's low and high case and its maximum absolute effect.

The run also writes the official UNFCCC extraction, the Climate TRACE fiscal and airport products, the route bounds, the completion ladder, the weather and observed-coverage pilots, the independence matrix, the QA table and a hash manifest.

The sector's own registered result is written by `make aviation` rather than by the study target, and its five tables were named in neither list above—not described and left nameless, simply absent, which is the harder omission to notice:

- `aviation_by_stream_fy2023_24.csv`: the registered stream ledger—category, subitem and gas with the fossil flag, gigagrams, both common-GWP vintages and the uncertainty per row.
- `aviation_fuel_constrained_by_stream_fy2023_24.csv`: the same ledger under the fuel constraint, which is the comparison the bridge exists to make.
- `aviation_provincial_fy2023_24.csv`: the province-level AR5 total.
- `aviation_monthly_fy2023_24.csv`: the twelve-month layer, each month's share with the emissions it implies in both vintages and the temporal method and basis behind the shape.
- `aviation_completeness_fy2023_24.csv`: one row per source stream with its status and the note that qualifies it.

14. Accounting and QA/QC guarantees

Automated tests require: OCAC JP-1 totals of 633 879 t and 658 334 t in the two fiscal years; exact closure of every source row; non-negative, non-overlapping assignments; monthly sums equal to their reported cumulative controls and any final-report revision remains explicit; calendar/fiscal benchmark periods; reproducible Monte Carlo draws; separate structural scenarios; route-bound ordering; identical common support; coverage completion to 20,300 stages; international memo exclusion; distinct LTO and full-flight bunker quantities; representative-engine provenance and imputation flags; and agreement between generated outputs and manuscript values.

The source hierarchy is IPCC Volume 2 Chapter 3 for greenhouse-gas boundaries and factors (IPCC, 2006); OCAC

Variable	Central	Range/distribution	Source or correlation group	Class
Official domestic movements	40,600	Reported; not sampled	PCAA FY2023–24	Parametric control
M0 aggregate activity/fleet representation	1.0	±60%; lognormal	Legacy M0 only	Parametric
Constrained route mean	486.5 nmi	485.2–503.0; bounded	PCAA margins/supported graph	Structural
M0 representative stage	550 nmi	±35%; lognormal	ICAO route sample; all stages	Parametric
Airline–aircraft shares	Route shares	Categorical draw	Fleet histories; operator–unordered route	Parametric within M2
OpenAP payload	PCAA passengers, cargo, mail	Observed-centred bounded triangular	Operator and aircraft family	Parametric
Route detour	1.08	±10%; lognormal	Shared reciprocal route	Parametric
Taxi-in/out	420/900 s	±50%; lognormal	Shared airport	Parametric
Unsupported-aircraft fuel	ICAO factor	OpenAP/ICAO; discrete	Aircraft family	Structural
Fuel carbon conversion	3.16 t/t	±5%; lognormal	IPCC/ICAO; all fuel	Parametric
CH4 factor	0.5 kg/TJ	±80%; lognormal	IPCC; all fuel	Parametric
N2O factor	2.0 kg/TJ	±110%; lognormal	IPCC; all fuel	Parametric
Customer to geography	OECD-calibrated	Literal/movement/carrier cases	OCAC/PCAA/OECD source row	Structural
International stage coverage	27.9%	Partial vs residual; discrete	Operator–route	Structural
Weather duration adjustment	Still air	ERA5 audit; empirical resampling	Global secondary interval	Parametric sensitivity

Table 4: Complete uncertainty-input register. The generated CSV additionally records the applicable model and exact source string. Reported controls are conditional on source quality; structural choices are not pooled into a confidence interval.

Stream/product	Status	Boundary note
Scheduled domestic full flight	Estimated	M0/M1/bridge; M2 diagnostic
International bunkers	Estimated	Full-flight bridge memo
International airport LTO	Estimated	Separate local-air product
General aviation/training	Partial	100LL priced as a memo row; activity unresolved
Military/government	Memo	JP-8/government candidates, priced as memo rows
Helicopters	Not estimated	No classified activity
APU/ground support	Not estimated	Airport survey required
Main-engine LTO pollutants	Estimated	Representative engines
Cruise pollutants/non-CO_2 effects	Not estimated	Separate research product
Tyre and brake wear	Not estimated	No local activity assignment

Table 5: Completeness matrix. Known omissions remain explicit rather than being absorbed into "non-commercial aviation".

for petroleum sales (Oil Companies Advisory Council, 2024); PCAA for movements and passengers (Pakistan Civil Aviation Authority, 2024b,a); Pakistan's BTR, NC3 and CRT for official national reporting (Government of Pakistan, 2024, 2025, 2026); OECD and Climate TRACE for external estimates (Garsous et al., 2022; Climate TRACE, 2026); ADSB.lol/readsb and ERA5 for the public coverage and weather pilots (ADSB.lol, 2024; readsb contributors, 2026; Copernicus Climate Change Service, 2026); ICAO/OpenAP for route- performance diagnostics (ICAO, 2024; Sun et al., 2020); ICAO Doc 9889 and the Engine Emissions Databank for airport pollutants (ICAO, 2023; ICAO and EASA, 2026); and the aviation chapter of EMEP/EEA for fuel-based pollutant factors (EMEP/EEA, 2023).

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